

6 Color Image Processing

- Color is a powerful description that often simplifies object identification and extraction from a scene
- Color Image Processing is divided into two major areas:
 - 1) Full-color processing
 - Images are acquired with a full-color sensor, such as a color video camera or color scanner
 - Color transformation, color complement and tone or color correction
 - 2) Pseudo color processing
 - Assigning a color to a particular monochrome intensity or range of intensities

RGB Color Models

- Model based on a Cartesian coordinate system
- Each color appears in its primary spectral components of red, green, and blue

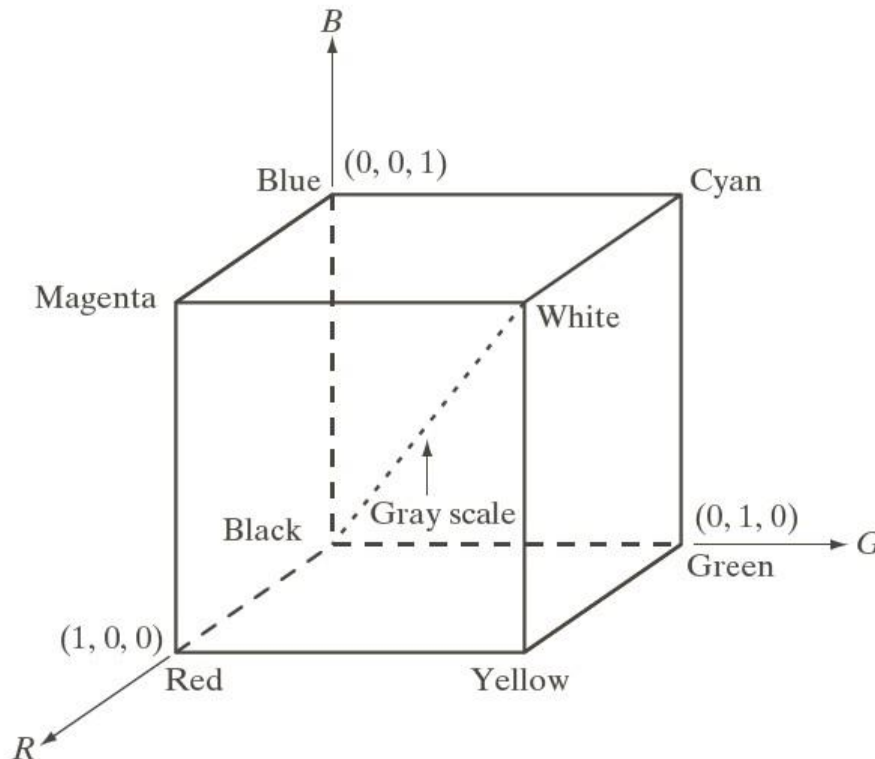


FIGURE 6.7

Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).

RGB Color Models

- RGB primary values are at three corners; the secondary colors cyan, magenta, and yellow are at the other corners; black is at the origin; and white is at the corner farthest from the origin
- In this model, the gray scale (points of equal RGB values) extends from black to white along the line joining these two points
- The different colors in this model are points on or inside the cube, and are defined by vectors extending from the origin

RGB Color Models

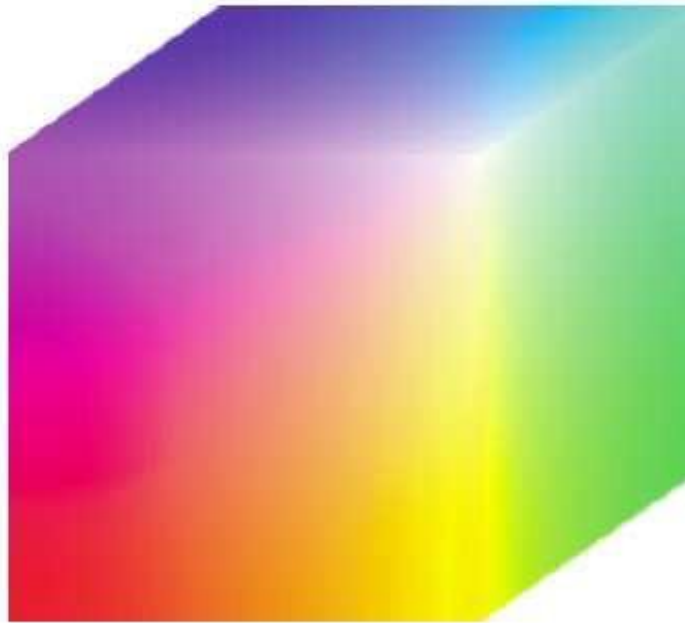


FIGURE 6.8 RGB
24-bit color cube.

- Number of bits used to represent each pixel in RGB space is called the *pixel depth*
- RGB image has 24 bit pixel depth
- True color or full color image is a 24 bit RGB image
- Total colors in 24-bit image is $(2^8)^3 = 16,777,216$